Md Mizanur Rahaman Nayan

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Research Interests

AI acceleration, Hardware–software co-design, Compute-in-memory, Emerging memory devices, AI infrastructure, VLSI, ASIC design, Energy-efficient computing, Explainable AI.

Education

Georgia Institute of Technology, Atlanta, GA *Aug 2023 – Present*
Ph.D. in Electrical and Computer Engineering

- Designed a SOT-CAM based vector symbolic macro for explainable AI; published at **ICCAD 2025**.
- Developed a systolic-array based accelerator with on-chip im2col for GeMM/Conv workloads; published at **DATE 2025**.
- Working on compute-in-memory architectures using SOT-MRAM for high-throughput ML kernels.
- Working on silicon tapeout project, Axon++, high throughput AI accelerator based on Axon.

Georgia Institute of Technology, Atlanta, GA *Aug 2023 – May 2025*
M.S. in Electrical and Computer Engineering

Bangladesh University of Engineering and Technology (BUET), Dhaka *Feb 2017 – May 2022*
B.Sc. in Electrical and Electronic Engineering

Industry Experience

Nissan Advanced Technology Center, Silicon Valley, CA May 2025 – Aug 2025
AI Hardware Accelerator Research Intern

- Profiled autonomous vehicles perception (camera/LiDAR) pipelines to identify compute, memory, and bandwidth bottlenecks across CPU, GPU, and RISC-V accelerator backends.
- RTL Designed and FPGA-prototype of a sparse convolution accelerator with RISCV-IMV, achieving 5× lower latency than NVIDIA RTX 3050.
- Built cycle-accurate performance and energy models, validated using simulators and FPGA.

ACI Limited, Dhaka, Bangladesh Jul 2022 – Aug 2023
Machine Learning Engineer

- Developed ML-based recommendation and customer recognition modules using Python and Django for enterprise-scale deployment. Insights from the workloads later used in AI accelerator co-design research.

Cobalt Speech and Language (Remote) Oct 2021 – Jan 2022
Machine Learning Research Intern

- Designed CycleGAN-based text normalization and speaker diarization pipelines; results published at **APSIPA ASC 2022**.

Selected Publications

- "**ACRONYM: Accelerated Approximate Nearest Neighbor Search in Memory for Dynamic Vector Databases**," MICRO 2026 [under review], **M. M. R. Nayan**, T. Zhang, F. Ponzina, T. Rosing, A. Naeemi.
- "**Cross-Layer Co-Designed In-Memory Acceleration for Real-Time Proteomics at the Edge**," ICCAD 2026 [under review], **M. M. R. Nayan**, Z. Li, F. Ponzina, S. Pinge, T. Rosing, A. Naeemi.
- "**Axon: A Novel Systolic Array Architecture for Improved Runtime and Energy-Efficient GeMM and Conv Operation with On-Chip im2col**," DATE 2025, **M. M. R. Nayan**, R. Raj, G. B. Shaik, T. Krishna, A. Naeemi.

- "**HyDra: SOT-CAM Based Vector Symbolic Macro for Hyperdimensional Computing**," ICCAD 2025, **M. M. R. Nayan**, C. K. Liu, Z. Wan, A. Raychowdhury, A. Naeemi.
- "**Frequency Tunable CMOS Ring Oscillator Based Ising Machine**," International Journal of Circuit Theory and Applications, 2024, **M. M. R. Nayan** and O. Hassan.
- "**Parallel Training of TN and ITN Models Through CycleGAN for Improved Sequence-to-Sequence Learning**," APSIPA ASC 2022, **M. M. R. Nayan** and M. A. Haque.
- "**A Deep Ensemble Model for Multi-Class Arrhythmia Classification Using 12-Lead ECG**," IEEE ICECE 2022, *S.Mahmud*[1], *M.M.R.Nayan*[1], *S.Hasan*[2], *M.N.Taj*[2].
- "**IoT-Based Real-Time Railway Fishplate Monitoring System for Early Warning**," IEEE ICECE 2020, *M.M.R.Nayan*[1], *S.A.Sufi*[2], *A.K.Abedin*[3], *R.Ahamed*[4] *and M.F. Hossain.*

Technical Skills

Programming: Python, C/C++, SystemVerilog, Bash, Assembly, SQL
ML/AI: PyTorch, TensorFlow, ONNX Runtime, TensorRT, Quantization, Profiling, Feature extraction
Hardware/EDA: Synopsys (VCS, DC, ICC2), Cadence Virtuoso, HSPICE, Verilator, Vivado, Vitis
Tools: VS Code, GitHub, Colab, MATLAB

Relevant Coursework

Advanced Computer Architecture; Advanced VLSI Systems; Hardware–Software Co-Design for ML; Parallel Programming for FPGAs; VLSI Theory to Tapeout; Advanced Programming Techniques

Honors and Awards

DAC Young Fellowship (2026); CRNCH Ph.D. Fellowship (2024); Silicon Creations ISSCC Grant (2024); Georgia Tech ECE Fellowship (2023); BUET Dean's List; 2nd Place — National Student-to-Startup Competition

Affiliations

Research Scholar, Semiconductor Research Corporation (SRC) — 2023 – Present
IEEE Student Member, Atlanta Section — 2023 – Present

Relevant Academic Projects

- **Silicon Tapeout** Designed a 4×4 systolic array and completed full ASIC flow (functional verification, synthesis and PnR). Built Python-based automated testbenches and implemented DFT using LFSR-based pattern generation and a signature analyzer wrapper. Taped out in TSMC 65GP node with 200 MHz frequency at an area constraint of $0.03mm^2$. Performed Ring Oscillator, Vdd vs Frequency (Shmoo Plot), and Current measurement, followed by functional validation of the chip.
- **Superscalar Processor (OoO + Branch Prediction)** Designed a superscalar OoO processor using the Tomasulo algorithm. Implemented multiple branch predictors (Always-Taken, Smith, Yeh-Patt, GShare), plus reservation stations, ROB, and register renaming; evaluated IPC gains.
- **Cache and Coherence Simulator** Built a cycle-accurate L1/L2 cache simulator with next-line and stride prefetchers. Implemented directory-based MESI coherence (agent + directory controllers) and analyzed miss rates and coherence traffic.
- **CNN Convolution Accelerator (on-the-fly im2col, Verilog)** Implemented an accelerator performing on-the-fly im2col with dilated-conv data feeding to reduce buffering and area. Verified design using Verilog testbenches and compared throughput to software baselines.
- **ASIC Design and Verification (Tic-Tac-Toe Processor)** Completed full RTL-to-GDSII flow: Verilog RTL, NC-SIM simulation, synthesis (Genus), and P&R + DRC/LVS (Innovus). Generated final GDSII and timing reports.
- **6T SRAM Cell Design and Write-Ability** Designed a 6T SRAM cell in 14nm PTM and evaluated write stability, SNM, and voltage sensitivity using Cadence Virtuoso/Spectre.